SANTHOSH SRINIVASAN

Data Engineer | Microsoft Fabric | Azure Databricks

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Professional Summary

Final-year B.Tech Computer Science and Business Systems student and Microsoft Certified Data Engineer with certifications in Azure Databricks and Microsoft Fabric. Hands-on experience building batch and streaming data pipelines using **Apache Spark, PySpark, Delta Lake, Microsoft Fabric, Azure Databricks, and Azure Data Factory**. Experienced in **ETL/ELT, incremental data processing, medallion architecture, dimensional modeling, and data warehousing**.

Technical Skills

Programming: Python, SQL
Data Engineering: ETL/ELT, Apache Spark, PySpark, Spark Structured Streaming, Delta Lake, Incremental Data Processing, Dimensional Data Modeling
Microsoft Fabric: Data Factory, Lakehouse, Warehouse, Notebooks, Data Pipelines
Databricks: Azure Databricks, Unity Catalog, Lakeflow Jobs, Lakeflow Declarative Pipelines (Spark Declarative Pipelines)
Azure: Azure Data Factory, Azure Data Lake Storage, Azure Synapse Analytics

Projects

RetailNova Data Engineering Case Study using Azure Databricks **Sep 2026**

- **Technologies:** Azure Databricks, PySpark, Delta Lake, Unity Catalog, SQL, Auto Loader, Lakehouse
- Built an end-to-end **incremental data engineering pipeline** using **Bronze, Silver, and Gold medallion architecture** to process **1M+ synthetic retail transaction records, 100K+ customers, and 10K+ products**.
- Implemented **incremental ingestion using Auto Loader and watermark-based processing**, processing newly arrived records across customer, product, and transaction datasets instead of reprocessing complete datasets.
- Developed PySpark transformations for **data cleansing, standardization, deduplication, null handling, and schema management** across Silver-layer datasets.
- Implemented **SCD Type-2** processing for customer data using **surrogate keys, effective dates, end dates, current-record flags, and incremental change detection** to preserve historical records.
- Implemented **SCD Type-1** processing for product data using **Delta Lake MERGE** to update existing records and insert new records.
- Applied **Delta Lake optimizations** using **Liquid Clustering, Optimized Writes, and Auto Compaction**, with **OPTIMIZE and VACUUM** for table maintenance.
- Designed a **Gold-layer dimensional model** using Delta tables and Unity Catalog for downstream analytics and reporting.

Car Sales Incremental Data Pipeline using Microsoft Fabric **Mar 2026**

- **Technologies:** Microsoft Fabric, Data Factory, PySpark, Delta Lake, Lakehouse, Warehouse, GitHub
- Built an end-to-end **incremental ELT pipeline** using Microsoft Fabric to ingest car sales data from GitHub into a staging database and process it through **Bronze, Silver, and Gold** layers.
- Implemented **watermark-based incremental loading** in Fabric Data Factory to identify load boundaries and process only newly arrived or changed records instead of reprocessing the full dataset.
- Developed PySpark transformations for **data cleansing and deduplication**, and implemented **SCD Type-1** dimension processing using Delta Lake MERGE to insert new records and update existing records.
- Designed a **Star Schema** with **dim_date, dim_branch, dim_model, dim_dealer, and fact_sales**, using surrogate keys and Delta tables in the Fabric Lakehouse for downstream analytics and Power BI consumption.

Real-Time Streaming Data Pipeline using Azure Databricks **Sep 2025**

- **Technologies:** Azure Databricks, Azure IoT Hub, Apache Spark, Spark Structured Streaming, Delta Lake
- Built an end-to-end **real-time streaming pipeline** to ingest simulated sensor data and process it using **Spark Structured Streaming**.
- Implemented fault-tolerant streaming with **Delta Lake checkpointing** and optimized storage using Optimize Write and Auto Compaction.

Education

B.Tech in Computer Science and Business Systems
Rajalakshmi Engineering College, Chennai | Anna University **2023 – 2027**

– CGPA: 8.38 / 10.0

Certifications

- Microsoft Certified: Azure Databricks Data Engineer Associate (DP-750)
- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Apache Airflow 3 Fundamentals
- Apache Airflow DAG Authoring
- DataCamp SQL Associate
- Databricks Platform Architect